

Elektrodynamik

Exam A Solutions

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PROF. N. BEISERT

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1. Short Questions

(20 points)

- a) Let A_μ be a constant four-vector and $B_\mu(x)$ be a four-vector field. Describe in words the transformation properties of the following quantities under Lorentz transformations: (2 points)

$$A^\mu A_\mu, \quad A^\mu B_\mu(x), \quad A_\mu A_\nu, \quad B_\mu(x) B_\mu(x). \quad (1)$$

Solution: The first quantity is a scalar, it is invariant. The second quantity is a scalar field. The third quantity is a (symmetric) covariant tensor of rank 2. The last quantity is neither covariant, contravariant nor invariant, it does not transform predictably under Lorentz boosts because the indices are not properly contracted, (1 point; for the first two correct answers) (1 point; if three answers are convincing).

- b) Derive the conservation equation relating the Poynting vector $\vec{S} = \mu_0^{-1}(\vec{E} \times \vec{B})$ and the energy density $w = \frac{1}{2}\varepsilon_0 \vec{E}^2 + \frac{1}{2}\mu_0^{-1} \vec{B}^2$ in free space. (4 points)

Solution: The Poynting vector serves as the energy flux density, and a continuity equation relates the time derivative of a density the divergence of the corresponding flux density (1 point). Up to a relative factor, the conservation equation should take the form (1 point*; for stating the form of the conservation equation up to a relative factor)

$$\vec{\partial} \cdot \vec{S} = -\partial_t w. \quad (S1)$$

In order to compute the divergence of the Poynting vector, we need the multiplication rule for the divergence of a cross product

$$\begin{aligned} \vec{\partial} \cdot (\vec{E} \times \vec{B}) &= \partial_i \varepsilon_{ijk} E_j B_k = B_k \varepsilon_{kij} \partial_i E_j - E_j \varepsilon_{jik} \partial_i B_k \\ &= \vec{B} \cdot (\vec{\partial} \times \vec{E}) - \vec{E} \cdot (\vec{\partial} \times \vec{B}). \end{aligned} \quad (S2)$$

We thus obtain (1 point; for deriving and/or applying the identity)

$$\vec{\partial} \cdot \vec{S} = \mu_0^{-1} \vec{B} \cdot (\vec{\partial} \times \vec{E}) - \mu_0^{-1} \vec{E} \cdot (\vec{\partial} \times \vec{B}). \quad (S3)$$

The two cross product terms can be simplified using two of the Maxwell equations at $\vec{j} = 0$ (1 point; for suitably applying at least one Maxwell equation)

$$\vec{\partial} \times \vec{E} = -\partial_t \vec{B}, \quad \vec{\partial} \times \vec{B} = \mu_0 \varepsilon_0 \partial_t \vec{E} + \mu_0 \vec{j} = \mu_0 \varepsilon_0 \partial_t \vec{E}. \quad (S4)$$

Substituting them in the divergence of the Poynting vector yields:

$$\vec{\partial} \cdot \vec{S} = -\mu_0^{-1} \vec{B} \cdot \partial_t \vec{B} - \varepsilon_0 \vec{E} \cdot \partial_t \vec{E}. \quad (S5)$$

The right-hand side yields the same terms (1 point; for evaluating the time derivative)

$$-\partial_t w = -\varepsilon_0 \vec{E} \cdot \partial_t \vec{E} - \mu_0^{-1} \vec{B} \cdot \partial_t \vec{B}. \quad (S6)$$

This shows that the factor between the two terms in the conservation equation is indeed as given above.

- c) Derive a formula for the velocity v of the Lorentz boost resulting from two consecutive Lorentz boosts with collinear velocities v_1 and v_2 . (2 points)

Solution: Without loss of generality we assume the Lorentz boosts to be along the x -axis (rotational symmetry). A Lorentz boost $\Lambda(v)$ is described by the matrix (the sign of v does not matter; and we can drop the y and z direction in which the boost acts trivially)

$$\Lambda(v) = \gamma(v) \begin{pmatrix} 1 & v/c \\ v/c & 1 \end{pmatrix}, \quad \gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (\text{S7})$$

The product of two transformations yields (1 point; for evaluating one element of the matrix product and demanding that it equals the corresponding element of the boost for the combined velocity)

$$\Lambda(v_1)\Lambda(v_2) = \gamma(v_1)\gamma(v_2) \begin{pmatrix} 1 & v_1/c \\ v_1/c & 1 \end{pmatrix} \begin{pmatrix} 1 & v_2/c \\ v_2/c & 1 \end{pmatrix} = \gamma(v_1)\gamma(v_2) \begin{pmatrix} 1 + v_1v_2/c^2 & (v_1 + v_2)/c \\ (v_1 + v_2)/c & 1 + v_1v_2/c^2 \end{pmatrix}. \quad (\text{S8})$$

Collinear Lorentz boosts form a group and thus two consecutive transformations correspond to a single total Lorentz boost $\Lambda(v) = \Lambda(v_1)\Lambda(v_2)$ with velocity v that is found from solving

$$\begin{aligned} \gamma(v) &= \gamma(v_1)\gamma(v_2)(1 + v_1v_2/c^2), \\ \gamma(v)v/c &= \gamma(v_1)\gamma(v_2)(v_1 + v_2)/c. \end{aligned} \quad (\text{S9})$$

This system is redundant and the solution for v is obtained immediately as the quotient of both equations (1 point; for stating the resulting law)

$$v = \frac{v_1 + v_2}{1 + v_1v_2/c^2}. \quad (\text{S10})$$

- d) Consider an inhomogeneous magnetic field $\vec{B}$ in free space. We place a magnetic dipole $\vec{m}$ at some position $\vec{x}$ and let it react freely to the field. Qualitatively describe the initial dynamics also stating the relevant direction(s). (3 points)

Solution: A dipole responds to an inhomogeneous magnetic field by a force and by a torque (1 point). The torque $\vec{m} \times \vec{B}(\vec{x})$ acts towards aligning the direction of the dipole with the direction of the magnetic field (1 point; for the formula or the qualitative direction). The force $\vec{\nabla}(\vec{m} \cdot \vec{B})$ pushes the dipole in the direction of the gradient of $\vec{m} \cdot \vec{B}$ (1 point; for the formula or the qualitative direction). Note that $\vec{m} \cdot \vec{B}$ represents a particular component of the magnetic field, not the magnetic field altogether.

- e) State the eigenvalue equations fulfilled by the spherical harmonic $Y_{\ell,m}$. (2 points)

Solution: The spherical harmonics fulfil the equations (2 points; one for each equation)

$$\vec{L}^2 Y_{\ell,m} = \ell(\ell + 1) Y_{\ell,m}, \quad L_z Y_{\ell,m} = m Y_{\ell,m}. \quad (\text{S11})$$

The solution can alternatively use the explicit expressions for $\vec{L}^2$ and L_z :

$$\begin{aligned} \vec{L}^2 &= -\frac{1}{\sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial}{\partial \vartheta} \right) - \frac{1}{\sin^2 \vartheta} \left(\frac{\partial}{\partial \varphi} \right)^2 \\ &= -\left(\frac{\partial}{\partial \vartheta} \right)^2 - \cot \vartheta \frac{\partial}{\partial \vartheta} - \frac{1}{\sin^2 \vartheta} \left(\frac{\partial}{\partial \varphi} \right)^2, \\ L_z &= -i \frac{\partial}{\partial \varphi}. \end{aligned} \quad (\text{S12})$$

- f) Consider a flat ring of inner and outer radii a , b . It carries a uniform surface charge density σ , and it rotates with angular velocity ω around its symmetry axis. Calculate the magnetic field $\vec{B}$ at the centre of the ring. (3 points)

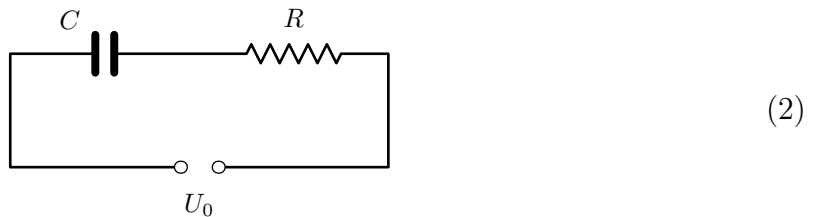
Solution: With $r = \|x\|$ and the ring located at the plane $z = 0$, the current density on the disc is (1 point; the distributional parts defining the disk may be skipped if treated consistently in the integral)

$$\vec{j}(x) = \sigma \vec{\omega} \times \vec{x} \delta(z) \theta(b-r) \theta(r-a) = \sigma \omega \vec{e}_z \times \vec{x} \delta(z) \theta(b-r) \theta(r-a) = \sigma \omega r \vec{e}_\varphi \delta(z) \theta(b-r) \theta(r-a). \quad (\text{S13})$$

The Biot–Savart law gives (1 point; for stating the Biot–Savart law) (1 point; for evaluating the result up to minor issues but including the direction)

$$\vec{B} = \int d^3x \frac{\mu_0 \vec{j}(x) \times (-\vec{x})}{4\pi r^3} = -\frac{\mu_0 \sigma \omega}{4\pi} \int d^3x \frac{\vec{e}_z r^2 - z \vec{x}}{r^3} \delta(z) = -\frac{1}{2} \mu_0 \sigma \omega \vec{e}_z \int_a^b dr = -\frac{1}{2} \mu_0 \sigma \omega (b-a) \vec{e}_z. \quad (\text{S14})$$

- g) Consider a circuit with a voltage source U_0 , a resistor of resistance R , and a capacitor of capacitance C .



Write down a differential equation for the charge on the capacitor, solve it assuming that the capacitor is initially uncharged, and calculate the time it takes to charge to 90% of its maximal charge. (4 points)

Solution: The voltage drop across the capacitor is determined by the definition of capacitance $U_C = Q/C$. The voltage drop across the resistors is given by Ohm's law $U_R = IR$, where we furthermore use $I = \dot{Q}$ (1 point; for the voltage drops of devices). According to Kirchhoff's first law, the sum of voltage drops on all components of a closed loop is 0. where a voltage drop U_0 of the source should be considered with negative sign along the loop (the sign is subject to interpretation, and can be flipped consistently). Alternatively, the sum of the voltage drops across the resistor and capacitor must equal the voltage supplied by the source Hence we get (1 point; for the composite relation)

$$\dot{Q}R + Q/C = U_0. \quad (\text{S15})$$

This is an inhomogeneous first-order differential equation. The homogeneous equation has the solution $Q_{\text{hom}} = K \exp(-t/RC)$, and a particular solution of the inhomogeneous equations is simply a constant $Q_0 = U_0 C$. The total solution is $Q(t) = Q_0 + Q_{\text{hom}}$. Since $Q(0) = U_0 C + K = 0$, we get the solution (1 point; for the homogeneous solution with correct physical sign in exponent and with a clear path towards the inhomogeneous solution)

$$Q(t) = U_0 C \left[1 - \exp\left(-\frac{t}{RC}\right) \right]. \quad (\text{S16})$$

As can be seen, the maximum (asymptotic) charge is Q_0 (but the capacitor will never fully charge to Q_0). It will be charged to 90% of Q_0 if (1 point; for solving the charging time)

$$1 - \exp\left(-\frac{t}{RC}\right) = 0.9 \quad \Rightarrow \quad \exp\left(-\frac{t}{RC}\right) = 0.1 \quad \Rightarrow \quad t = RC \log(10). \quad (\text{S17})$$

2. Covariant Plane Waves (18 points)

In this problem, we consider the relativistic formulation of an electromagnetic monochromatic plane wave in empty space.

- a) Find a gauge condition for the four-potential A_μ such that the wave equation reduces to $\square A_\mu = 0$. Describe briefly a relevant advantage of this gauge. (3 points)

Solution: We start with the inhomogeneous Maxwell equation in vacuum: (1 point; for ME, substitution and expansion)

$$0 = \mu_0 j^\nu = \partial_\mu F^{\mu\nu} = \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = \partial^2 A^\nu - \partial^\nu \partial_\mu A^\mu. \quad (\text{S18})$$

Upon fixing Lorenz gauge (1 point; for gauge condition or its name)

$$\partial_\mu A^\mu = 0, \quad (\text{S19})$$

we obtain

$$0 = \partial^2 A^\nu = \square A^\nu. \quad (\text{S20})$$

The Lorenz gauge is a useful gauge choice as it is Lorentz in-/covariant and it decouples the components of the wave equation (1 point; for either feature). One can argue that indices in (S20) are properly contracted, and thus this gauge choice is compatible with Lorentz symmetry.

In fact, one might generalise the Lorenz gauge condition to

$$\partial_\mu A^\mu = \text{const}. \quad (\text{S21})$$

Here, const can either represent an overall fixed constant α or merely the requirement that $\partial_\mu A^\mu$ does not depend on spacetime with an arbitrary value of α . These extensions of the Lorenz gauge are perfectly valid answers. Note that the subsequent parts on monochromatic plane waves effectively work in Fourier space where the constant term represents a distribution $\delta^4(k)$. Thus they lead to no change in the subsequent parts except for the special case $k_\nu = 0$ which is too special to require dedicated attention.

We will remain in this gauge throughout this problem and we will consider a monochromatic plane wave solution for the four-potential A_μ of the form

$$A_\mu(x) = \text{Re}[a_\mu e^{ik_\nu x^\nu}], \quad (3)$$

where a_μ and k_ν are complex and real four-vectors, respectively.

- b) Derive suitable conditions for a_μ and k_ν such that A_μ solves the wave equation. Briefly interpret these relations. (3 points)

Solution: A solution must respect the wave equation $\square A^\nu = \partial_\mu \partial^\mu A^\nu = 0$ and the Lorenz gauge condition $\partial_\mu A^\mu = 0$. Due to linearity, the projection to real parts may be skipped.

The ansatz (3) satisfies the wave equation when (1 point)

$$\partial_\mu \partial^\mu A_\nu = \text{Re}[-k_\mu k^\mu a_\nu e^{ik \cdot x}] = 0 \quad \Longleftrightarrow \quad k_\mu k^\mu = 0. \quad (\text{S22})$$

This is the vacuum dispersion relation for light requiring the wave four-vector to be null.

The ansatz (3) respects the Lorenz gauge condition when (1 point)

$$\partial_\mu A^\mu = \text{Re}[-ik_\mu a^\mu e^{ik \cdot x}] = 0 \quad \Longleftrightarrow \quad k_\mu a^\mu = 0. \quad (\text{S23})$$

This is the transversality condition showing that the polarisation of the wave is perpendicular to the direction of propagation (1 point; for interpretations of $k_\mu k^\mu = 0$ and $k_\mu a^\mu = 0$, mainly the former).

- c) Describe the residual gauge freedom in this gauge. What ambiguity does this imply for the amplitude a_μ in the monochromatic plane wave ansatz (3)? How many physical degrees of freedom does a_μ support? (4 points)

Solution: Under a gauge transformation, the four-potential transforms as

$$A'_\mu = A_\mu + \partial_\mu \Lambda. \quad (\text{S24})$$

The Lorenz gauge $\partial_\mu A^\mu = 0$ is thus preserved by a gauge transformation Λ that satisfies

$$\partial_\mu A'^\mu = \partial_\mu A^\mu + \partial_\mu \partial^\mu \Lambda = \square \Lambda = 0. \quad (\text{S25})$$

As this equation has non-trivial solutions, the Lorenz gauge is only a partial gauge, i.e. it only fixes the gauge potential A_μ partially. (1 point; for a qualitative statement).

In particular, one can solve the wave equation for Λ as a superposition of monochromatic plane waves (1 point; for a precise description)

$$\Lambda = \text{Re}[\lambda e^{ik \cdot x}] \quad (\text{S26})$$

with amplitude λ and null wave vector k_μ . In the monochromatic and linear setup, we need to consider only such gauge transformations with the same wave vector as the one for the plane wave solution. The four-potential gauge transforms as

$$A'_\mu = \text{Re}[a_\mu e^{ik \cdot x} + i\lambda k_\mu e^{ik \cdot x}] = \text{Re}[(a_\mu + i\lambda k_\mu) e^{ik \cdot x}], \quad a'_\mu = a_\mu + i\lambda k_\mu. \quad (\text{S27})$$

Consequently, the amplitude vector a_μ is defined only modulo shifts by the wave vector k_μ (1 point; for the transformation of a_μ). Any two amplitude vectors differing by a (complex) multiple of the wave vector yield the same field strength tensor and are thus physically equivalent. The amplitude a_μ has 4 (complex) degrees of freedom, one is fixed by the transversality condition and another one is the above ambiguity. Hence there are two physical degrees of freedom (1 point). Alternatively, one may recall that monochromatic light has two polarisation modes (1 point*).

d) Compute the field strength tensor $F_{\mu\nu}$ and show that it satisfies the Maxwell equations. (3 points)

Solution: The resulting field strength tensor reads (1 point)

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = \text{Re}[(ik_\mu a_\nu - ik_\nu a_\mu) e^{ik \cdot x}] = \text{Im}[(a_\mu k_\nu - a_\nu k_\mu) e^{ik \cdot x}]. \quad (\text{S28})$$

We may drop the projection to the real or imaginary part. For the inhomogeneous Maxwell equations $\partial^\mu F_{\mu\nu} = 0$ we obtain the relation

$$k^\mu (a_\mu k_\nu - a_\nu k_\mu) = (k \cdot a) k_\nu - k^2 a_\nu = 0 \quad (\text{S29})$$

These are satisfied thanks to the relations derived in part b) (1 point). If the relations $k \cdot a = k^2 = 0$ are derived here, the corresponding points may be awarded in part b).

The homogeneous equations $\varepsilon_{\mu\nu\rho\sigma} \partial^\nu F^{\rho\sigma} = 0$ are automatically satisfied because the field strength has been derived from a potential (1 point). Alternatively, one can compute explicitly

$$\varepsilon_{\mu\nu\rho\sigma} k^\nu a^\rho k^\sigma - \varepsilon_{\mu\nu\rho\sigma} k^\nu k^\rho a^\sigma = -2\varepsilon_{\mu\nu\rho\sigma} k^\nu k^\rho a^\sigma = 0, \quad (\text{S30})$$

which identically vanishes because $\varepsilon_{\mu\nu\rho\sigma}$ is anti-symmetric in the pair of indices (ν, ρ) while $k^\nu k^\rho a^\sigma$ is symmetric in these indices (1 point*).

e) Show that the electromagnetic fields $\vec{E}$ and $\vec{B}$ derived from the field strength tensor $F_{\mu\nu}$ together with the wave vector $\vec{k}$ form an orthogonal triad. (5 points)

Solution: The electric and magnetic fields are given by (1 point; factors of proportionality do not matter much)

$$E_k = cF_{0k}, \quad B_k = -\frac{1}{2}\varepsilon_{ijk}F_{ij}. \quad (\text{S31})$$

The orthogonality of $\vec{E}$ and $\vec{k}$ follows directly from the inhomogeneous Maxwell equations (1 point)

$$\vec{k} \cdot \vec{E} \sim \vec{\partial} \cdot \vec{E} = c\partial^\mu F_{0\mu} = 0. \quad (\text{S32})$$

The orthogonality of $\vec{B}$ and $\vec{k}$ follows directly from the homogeneous Maxwell equations (1 point)

$$\vec{k} \cdot \vec{B} \sim \vec{\partial} \cdot \vec{B} = -\frac{1}{2} \varepsilon_{kij} \partial_k F_{ij} = -\frac{1}{2} \varepsilon_{0\nu\rho\sigma} \partial^\nu F^{\rho\sigma} = 0. \quad (\text{S33})$$

Alternatively, one might refer to the two scalar Maxwell equations in vector notation (2 points*).

The orthogonality of $\vec{E}$ and $\vec{B}$ follows from the remaining (in)homogeneous Maxwell equations. For example, consider the remaining homogeneous equations

$$0 = \varepsilon_{k\nu\rho\sigma} \partial^\nu F^{\rho\sigma} \sim \varepsilon_{k\nu\rho\sigma} k^\nu F^{\rho\sigma} = -\varepsilon_{0kij} \varepsilon_{ijl} c^{-1} k_0 B_l - 2c^{-1} \varepsilon_{0kij} k_i E_j = -2c^{-1} k_0 B_k - 2c^{-1} \varepsilon_{kij} k_i E_j. \quad (\text{S34})$$

This is equivalent to (1 point; for deriving or simply stating this or the opposite relation)

$$\vec{B} = -\frac{1}{k_0} \vec{k} \times \vec{E}, \quad (\text{S35})$$

which implies that $\vec{B}$ is orthogonal to $\vec{E}$ (1 point) and $\vec{k}$ (1 point*; for orthogonality with $\vec{k}$).

3. Conducting Half-Space (20 points)

Consider a setup where the half-space at $z \leq 0$ is filled with perfectly conducting matter. The complementary half-space at $z > 0$ is empty except for a static charge distribution described by the charge distribution function

$$\rho^0(x, y, z) := C \delta(x) \delta(y) \theta(z - A), \quad A > 0, \quad (4)$$

where $\theta(z)$ denotes the unit step function with $\theta(-x) = 0$ and $\theta(+x) = 1$ for $x > 0$.

- a) Describe the geometry and orientation of the given charge distribution as well as the quantity C in words. (3 points)
-

Solution: The charge distribution describes a homogeneous straight linear charge distribution (1 point; for homogeneous and/or straight) with linear charge density C (1 point; for linear charge density) which is orthogonal and at a distance of A to the surface of the conductor (1 point; for orientation and/or distance).

- b) Compute the electric field $\vec{E}^0$ induced by the distribution ρ^0 assuming that the conducting material at $z \leq 0$ is absent. (6 points)

Hint: One of the following two derivative identities may be of help:

$$\frac{\partial}{\partial x} \frac{x}{\sqrt{x^2 + B}} = \frac{B}{(x^2 + B)^{3/2}}, \quad \frac{\partial}{\partial x} \log \frac{\sqrt{x^2 + B} + x}{D} = \frac{1}{\sqrt{x^2 + B}}. \quad (5)$$

Solution: The linear charge distribution induces the electric field: (1 point; for the integral expression)

$$\vec{E}^0 = \int_A^\infty \frac{C}{4\pi\varepsilon_0} \frac{dz' (x, y, z - z')}{(x^2 + y^2 + (z - z')^2)^{3/2}}. \quad (\text{S36})$$

For the longitudinal component E_z^0 we use that the integrand has the form $f'/f^{3/2}$ with $f(z') = x^2 + y^2 + (z - z')^2$. The indefinite integral yields $-2/f^{1/2}$ (1 point; for the indefinite integral). We thus find: (1 point; for the evaluation at $z' = A$)

$$E_z^0 = -\frac{C}{4\pi\varepsilon_0} \frac{1}{\sqrt{x^2 + y^2 + (z - A)^2}}. \quad (\text{S37})$$

For the transverse components $E_x^0 = xI$ and $E_y^0 = yI$ we can integrate the integrand of I using the first hint (5) with $B = x^2 + y^2$ and dividing by B : (1 point; for the indefinite integral) (1 point; for the evaluation at $z' = A$) (1 point; for the evaluation at $z' = \infty$)

$$I = \int_A^\infty \frac{C}{4\pi\epsilon_0} \frac{dz'}{(x^2 + y^2 + (z - z')^2)^{3/2}} = \frac{C}{4\pi\epsilon_0(x^2 + y^2)} \left[1 + \frac{z - A}{\sqrt{x^2 + y^2 + (z - A)^2}} \right]. \quad (\text{S38})$$

Alternatively, we can consider the electrostatic potential (1 point*; for the integral expression)

$$\Phi = \int_A^\infty \frac{C}{4\pi\epsilon_0} \frac{dz'}{\sqrt{x^2 + y^2 + (z - z')^2}}. \quad (\text{S39})$$

The integrand can be integrated using the second hint (5) with $B = x^2 + y^2$: (1 point*; for the indefinite integral) (1 point*; for the evaluation at $z' = A$)

$$\Phi = -\frac{C}{4\pi\epsilon_0} \log \frac{\sqrt{x^2 + y^2 + (z - A)^2} - z + A}{D}. \quad (\text{S40})$$

Note, however, that the integral does not converge, but the logarithmic divergence has a constant spatial profile and such a constant contribution to the potential has no physical significance (1 point*; for noticing a convergence issue or stating the arbitrariness or irrelevance of a constant).

The gradient of the potential determines the electric field (1 point*; for the x -component) (1 point*; for the z -component)

$$\begin{aligned} E_x^0 &= -\partial_x \Phi = \frac{C}{4\pi\epsilon_0} \frac{x}{\sqrt{x^2 + y^2 + (z - A)^2}} - z + A \frac{x}{\sqrt{x^2 + y^2 + (z - A)^2}} \\ &= \frac{C}{4\pi\epsilon_0} \frac{x}{x^2 + y^2} \left[1 + \frac{z - A}{\sqrt{x^2 + y^2 + (z - A)^2}} \right], \\ E_y^0 &= \dots = (y/x)E_x, \\ E_z^0 &= -\partial_z \Phi = \frac{C}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2 + y^2 + (z - A)^2} - z + A} \left[\frac{z - A}{\sqrt{x^2 + y^2 + (z - A)^2}} - 1 \right] \\ &= -\frac{C}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2 + y^2 + (z - A)^2}}. \end{aligned} \quad (\text{S41})$$

- c) Explain why the field $\vec{E}^0$ obtained in part b) cannot be the complete result when the conductor is present. Determine the response field $\vec{E}^R$ which is induced by the shifting of charges within the conducting half-space, both for $z > 0$ and $z < 0$. Then show explicitly that the total field $\vec{E} = \vec{E}^0 + \vec{E}^R$ has the desired properties. (6 points)

Hint: It suffices to express the field $\vec{E}^R(x, y, z)$ in terms of the field $\vec{E}^0(x, y, z)$.

Solution: The field $\vec{E}^0$ does not respect the boundary condition $\vec{E}_\parallel^0 = 0$ for surfaces of conductors. Furthermore, it does not meet the bulk criterion $\vec{E}^0 = 0$ for conductors in electrostatics (1 point; for either violation). Alternatively, one can state that the potential Φ^0 is constant neither on the interface nor in the bulk (1 point*; for either violation).

The setup of a conducting half-space can be solved by mirror charges. The mirror charges induce a response vector field for $z > 0$ which is mirrored at the surface (1 point; for mirrored coordinate dependence) and has the direction inverted (1 point; for the parallel vector components properly mirrored). The response field for $z > 0$ can be expressed as:

$$\begin{aligned} E_x^R(x, y, z) &= -E_x^0(x, y, -z), \\ E_y^R(x, y, z) &= -E_y^0(x, y, -z), \\ E_z^R(x, y, z) &= E_z^0(x, y, -z). \end{aligned} \quad (\text{S42})$$

Equivalently, the response potential for $z > 0$ is the negative (1 point*; for flipped sign) of the mirrored potential (1 point*; for mirrored coordinate dependence) (up to a constant)

$$\Phi^R(x, y, z) = -\Phi^0(x, y, -z) + \text{const.} \quad (\text{S43})$$

The total field (potential) on the interface $z = 0$ reads (1 point; for evaluation on the boundary $z = 0$ and simplifying)

$$\begin{aligned} E_x(x, y, 0) &= E_x^0(x, y, 0) - E_x^0(x, y, 0) = 0, \\ E_y(x, y, 0) &= E_y^0(x, y, 0) - E_y^0(x, y, 0) = 0, \\ E_z(x, y, 0) &= E_x^0(x, y, 0) + E_z^0(x, y, 0) = 2E_z^0(x, y, 0), \\ \Phi(x, y, 0) &= \Phi(x, y, 0) - \Phi(x, y, 0) + \text{const} = \text{const.} \end{aligned} \quad (\text{S44})$$

This confirms the appropriate boundary condition $E_{\parallel} = 0$ or $\Phi = \text{const}$ (1 point; for correctly showing it).

Furthermore, the response field (potential) within the conducting half-space $z < 0$ is just the negative of the original field at the same location:

$$\begin{aligned} \vec{E}^R(x, y, z) &= -\vec{E}^0(x, y, z), \\ \Phi^R(x, y, z) &= -\Phi^0(x, y, z). \end{aligned} \quad (\text{S45})$$

Therefore, the field (potential) within the conducting half-space $z < 0$ is zero (constant) as it should (1 point; for either of the above statements concerning $z < 0$).

- d) Describe two different charge densities ρ^R and $\tilde{\rho}^R$ within the conducting half-space $z \leq 0$ that equivalently induce the response field $\vec{E}^R$ for $z > 0$ obtained in part c) (both in words as well as in mathematical terms). Does the total of the induced charge on the conductor match your expectations? Why (not)? (5 points)

Solution: Two solutions are mirror charges $\tilde{\rho}^R$ (1 point; for mentioning the mirror charge density) and the actual induced surface charge density $\rho^R = \delta(z)\sigma^R$ on the interface (1 point; for mentioning or describing a surface charge density). The former is given by the negative mirror image of the original charge distribution ρ^0 (1 point; for the mirror charge density)

$$\tilde{\rho}^R(x, y, z) = -\rho(x, y, -z) = -C\delta(x)\delta(y)\theta(-z - A). \quad (\text{S46})$$

The latter is given by the discontinuity of the normal component of the electric field at the interface (1 point; for the surface charge density; unless it is zero due to earlier sign mistake for E_z^R)

$$\sigma^R(x, y, 0) = \epsilon_0 E_z(x, y, 0^+) - \epsilon_0 E_z(x, y, 0^-) = 2\epsilon_0 E_z^0(x, y, 0) - 0 = 2\epsilon_0 E_z^0(x, y, 0). \quad (\text{S47})$$

The total induced charge on the conducting half-space is the negative of the total charge on the outside. Equivalently, one can say that the surface charge density is everywhere negative (in fact, the total charge is infinite, but clearly non-zero) (0 points; for the factual statement). This is expected in general for the planar mirror charge setup. One may also expect that the total induced charge on a (connected) conductor is zero (electrons and protons as charge carriers will not simply appear or disappear). The expectation does not hold because the conductor is non-compact, and thus some charges can escape to infinity where they escape the charge balance. In physical reality the expectation of neutrality nevertheless does hold because no physical conductor has infinite extent. (1 point; for any applicable reason that goes beyond a factual statement about the situation).

4. Shielding of Currents (24 points)

In this problem, we study magnetic shielding for a pair of opposing currents. First, consider two infinitely long straight wires parallel to the z -axis. These two wires at positions $(x, y) = (\pm a, 0)$ carry the currents $\pm I$ in the positive z -direction.

- a) Compute the magnetic field $\vec{H}$ induced by the currents and express the result in Cartesian coordinates x, y, z . (3 points)

Solution: The magnetic field is given by the superposition of the magnetic fields of the two wires. The magnetic field of a single wire at $(x, y) = (0, 0)$ with current I in the positive z -direction takes the form (1 point; for rational function) (1 point; for vector part)

$$\vec{H} = \frac{I\vec{e}_z \times \vec{x}}{2\pi\|\vec{x}\|^2} = \frac{I}{2\pi} \frac{-y\vec{e}_x + x\vec{e}_y}{x^2 + y^2}. \quad (\text{S48})$$

The combination of the two wires therefore yields (1 point; for superposition)

$$\begin{aligned} \vec{H} &= \frac{(I\vec{e}_z) \times (\vec{x} - a\vec{e}_x)}{2\pi\|\vec{x} - a\vec{e}_x\|^2} + \frac{(-I\vec{e}_z) \times (\vec{x} + a\vec{e}_x)}{2\pi\|\vec{x} + a\vec{e}_x\|^2} \\ &= \frac{I}{2\pi} \frac{-y\vec{e}_x + (x-a)\vec{e}_y}{(x-a)^2 + y^2} - \frac{I}{2\pi} \frac{-y\vec{e}_x + (x+a)\vec{e}_y}{(x+a)^2 + y^2}. \end{aligned} \quad (\text{S49})$$

This expression may be combined as (0 points; combination may be awarded towards part b))

$$\vec{H} = -\frac{I}{2\pi} \frac{2a(2xy\vec{e}_x + (a^2 + y^2 - x^2)\vec{e}_y)}{(x^2 - 2ax + a^2 + y^2)(x^2 + 2ax + a^2 + y^2)}. \quad (\text{S50})$$

- b) Show that the magnetic field of the two currents can be represented as $\vec{H} = -\vec{\partial}\Psi$ by the scalar magnetic potential

$$\Psi = -\frac{I}{2\pi} \arctan\left(\frac{2ay}{x^2 + y^2 - a^2}\right). \quad (6)$$

Briefly explain why a scalar magnetic potential may be used in this case, as well as the shortcomings of this description. (5 points)

Hint: $(\partial/\partial z) \arctan z = 1/(1+z^2)$.

Solution: We compute the gradient as (2 points)

$$\begin{aligned} H_x &= -\frac{\partial\Psi}{\partial x} = -\frac{I}{2\pi} \frac{4axy}{(x^2 + y^2 - a^2)^2 + 4a^2y^2}, \\ H_y &= -\frac{\partial\Psi}{\partial y} = \frac{I}{2\pi} \frac{2a(x^2 + y^2 - a^2) - 4ay^2}{(x^2 + y^2 - a^2)^2 + 4a^2y^2} = \frac{I}{2\pi} \frac{2a(x^2 - y^2 - a^2)}{(x^2 + y^2 - a^2)^2 + 4a^2y^2}. \end{aligned} \quad (\text{S51})$$

Noting the identity

$$(x^2 + y^2 - a^2)^2 + 4a^2y^2 = (x^2 - 2ax + a^2 + y^2)(x^2 + 2ax + a^2 + y^2) \quad (\text{S52})$$

the magnetic field agrees with (S49) (1 point; for combining the terms and comparing).

Alternatively, especially if the comparison to the results of part a) does not work out, one may argue that that Laplace equation should hold for Ψ almost everywhere (1 point*; for conveying some understanding) and subsequently show that it does (2 points*; for showing harmonicity at generic points).

The current density sourcing the magnetic field is zero almost everywhere. Therefore, the magnetic field is curl-free $\vec{\partial} \times \vec{H} = 0$ almost everywhere (1 point; for mentioning absence of sources, absence of current density or curl-free). and its may be expressed as the gradient of a scalar field on a simply connected domain. For the given configuration the domain has punctures at $(x, y) = (\pm a, 0)$ and is thus not simply connected. This leads to a potential which is multi-valued, as expressed by the ambiguity of the arctan by shifts of integer multiples of π (1 point; for mentioning simply connected domain or multi-valued).

For the remainder of this problem, we consider the “current dipole limit” where we bring the wires close together, $a \rightarrow 0$, while making the current large, $I \rightarrow \infty$, such that the current dipole moment $\eta := 2aI$ remains finite.

c) Compute the dipole limit of the scalar magnetic potential Ψ . (1 point)

Solution: In the limit, the argument of arctan becomes small and we can approximate $\arctan z$ by the leading linear term z . We thus find (1 point)

$$\Psi \rightarrow -\frac{I}{2\pi} \frac{2ay}{x^2 + y^2} = -\frac{\eta}{2\pi} \frac{y}{x^2 + y^2} = -\frac{\eta}{2\pi r} \sin(\varphi). \quad (\text{S53})$$

The wires are now embedded in a long circular cylinder of radius R along the z -axis centred at $(x, y) = (0, 0)$. The cylinder is filled with matter of relative magnetic permeability μ_r with free space on the outside.

d) Explain briefly using symmetry arguments why we can make the ansatz $\Psi(r, \varphi, z) = P(r) \sin(\varphi)$ for the configuration using cylindrical coordinates r, φ, z . (3 points)

Solution: The configuration has cylindrical symmetry consisting of rotations around the axis and translations along the axis. Translational symmetry ensures that the z -dependency will be respected by all equations and boundary conditions. Here, the z -dependency of the induced potential is trivial (1 point). In fact, the displacement of the wires breaks the rotational symmetry (1 point; for broken rotational symmetry), but only to the extent that the source terms $\sim e^{\pm i\varphi}$ acquire well-prescribed phases. The currents induce a potential $\Psi \sim \sin(\varphi)$ and due to rotational symmetry the φ -dependency will be respected by all equations and boundary conditions. Consequently, only terms with $m = \pm 1$ may/should contribute, more concretely only those with $\sim e^{i\varphi} - e^{-i\varphi} \sim \sin(\varphi)$ as can be argued using parity (1 point). This explains the applicability of the ansatz $\Psi(r, \varphi, z) = P(r) \sin(\varphi)$.

e) Argue further that a suitable ansatz for the function $P(r)$ takes the form

$$P(r) = \begin{cases} Ar + B/r & \text{for } r < R, \\ Cr + D/r & \text{for } r > R, \end{cases} \quad (7)$$

with some constants A, B, C, D left to be determined. (4 points)

Hint: The Laplace operator can be expressed as $\Delta = \partial_r^2 + r^{-1}\partial_r + r^{-2}\partial_\varphi^2 + \partial_z^2$.

Solution: The magnetic field and thus the potential is source-free, $\Delta\Psi = 0$, everywhere except for the radii $r = 0$ and $r = R$. We apply the Laplace operator to the ansatz and find the differential equation (1 point)

$$\frac{\Delta\Psi}{\Psi} = \frac{P''(r)}{P(r)} + \frac{P'(r)}{rP(r)} - \frac{1}{r^2}. \quad (\text{S54})$$

As the equation has homogeneous degree -2 in r , we can make the monomial ansatz $P(r) = r^n$ (1 point) to find

$$\frac{\Delta\Psi}{\Psi} = \frac{n(n-1) + n - 1}{r^2} = \frac{n^2 - 1}{r^2} \stackrel{!}{=} 0. \quad (\text{S55})$$

The relevant source-free potentials are thus given by $n = \pm 1$ (1 point). It also suffices to show that both $r^{\pm 1} \sin(\varphi)$ satisfy the Laplace equation which should have a basis of two solutions.

These solutions apply within the intervals $0 < r < R$ and $r > R$ whereas $P(r)$ will not be smooth at the interface $r = R$ (1 point).

f) Use appropriate asymptotics at $r \rightarrow \infty$ as well as at $r \rightarrow 0$ to determine two of the constants. (2 points)

Solution: Inside the cylinder we should match with the singular behaviour of the previously determined potential. Therefore, we can set $B = -\eta/2\pi$ (1 point). The other solution $\sim r$ does not display singular behaviour at $r \rightarrow 0$ and the corresponding coefficient A is determined by something else.

Furthermore, we assume that the magnetic field vanishes at $r \rightarrow \infty$. The coefficient C corresponds to an asymptotically constant magnetic field, thus we should set $C = 0$ (1 point). Even though it is not warranted by the setup of the problem, C may also be left unfixed (1 point*; with explanation).

- g) Use the magnetostatic continuity conditions at $r = R$ to establish and solve two equations that determine the remaining two constants. What can you conclude about the shielding effect of the material? (6 points)

Solution: The magnetostatic continuity conditions require that $H_{\parallel} = H_{\varphi}$ and $B_{\perp} = B_r$ are continuous at $r = R$ (2 points). In our ansatz $\vec{H} = \vec{\partial}\Psi$, these are given by (2 points)

$$H_{\varphi} = -\frac{1}{r} \frac{\partial \Psi}{\partial \varphi} = -\frac{P(r)}{r} \cos(\varphi), \quad B_r = -\mu(r) \frac{\partial \Psi}{\partial r} = -\mu(r) P'(r) \sin(\varphi). \quad (\text{S56})$$

Effectively, the constraints impose that the functions $P(r)$ and $\mu(r)P'(r)$ are continuous at the interface $r = R$. We compute

$$P(r) = \begin{cases} Ar + B/r & \text{for } r < R, \\ Cr + D/r & \text{for } r > R, \end{cases} \quad \mu(r)P'(r) = \begin{cases} \mu_0\mu_r(A - B/r^2) & \text{for } r < R, \\ \mu_0(C - D/r^2) & \text{for } r > R. \end{cases} \quad (\text{S57})$$

Matching these equations at the interface $r = R$ results in

$$AR - \frac{\eta}{2\pi R} = \frac{D}{R}, \quad \mu_r \left(A + \frac{\eta}{2\pi R^2} \right) = -\frac{D}{R^2}. \quad (\text{S58})$$

Altogether, the solution is given by (1 point)

$$A = -\frac{\eta}{2\pi R^2} \frac{\mu_r - 1}{\mu_r + 1}, \quad B = -\frac{\eta}{2\pi}, \quad C = 0, \quad D = -\frac{\eta}{2\pi} \frac{2\mu_r}{\mu_r + 1}. \quad (\text{S59})$$

or

$$P(r) = -\frac{\eta}{2\pi} \begin{cases} \frac{1}{r} + \frac{\mu_r - 1}{\mu_r + 1} \frac{r}{R^2} & \text{for } r < R, \\ \frac{2\mu_r}{\mu_r + 1} \frac{1}{r} & \text{for } r > R. \end{cases} \quad (\text{S60})$$

Across the interface, the asymptotic magnetic field reduces by a factor of $D/B = 2\mu_r/(\mu_r + 1)$ (1 point).

5. Waves in a Topological Insulator (20 points)

In this problem, we consider a topological insulator which has some remarkable optical properties. With a constant α , the defining relations for the material take the form:

$$\vec{D} = \varepsilon \vec{E} - \sqrt{\varepsilon/\mu} \alpha \vec{B} \quad \text{and} \quad \vec{H} = \mu^{-1} \vec{B} + \sqrt{\varepsilon/\mu} \alpha \vec{E}. \quad (8)$$

- a) Express the free Maxwell equations and the defining relations for the topological insulator for a monochromatic plane wave ansatz for the electromagnetic fields $\vec{E}, \vec{D}, \vec{B}, \vec{H}$. Reduce and express the resulting relations in terms of $\vec{E}$ and $\vec{B}$, and determine the velocity v of the waves. (6 points)

Hint: Complexify the fields and assume a definite wave number and angular frequency.

Solution: For a monochromatic plane wave with spatial and temporal dependency $\exp(i\vec{k} \cdot \vec{r} - i\omega t)$, the four Maxwell equations in matter read (2 points)

$$\vec{k} \cdot \vec{B} = 0, \quad \vec{k} \times \vec{E} = \omega \vec{B}, \quad \vec{k} \cdot \vec{D} = 0, \quad \vec{k} \times \vec{H} = -\omega \vec{D}. \quad (\text{S61})$$

Inserting the defining relations, the third equation together with the first yields (1 point)

$$0 = \vec{k} \cdot \vec{D} = \varepsilon \vec{k} \cdot \vec{E} - \sqrt{\varepsilon/\mu} \alpha \vec{k} \cdot \vec{B} \quad \implies \quad \vec{k} \cdot \vec{E} = 0. \quad (\text{S62})$$

We can also contract the second equation with $\vec{E}$ to find that $\vec{E}, \vec{B}$ and $\vec{k}$ are orthogonal as usual for plane monochromatic waves.

The fourth equation together with the defining relations yields

$$\mu^{-1} \vec{k} \times \vec{B} + \sqrt{\varepsilon/\mu} \alpha \vec{k} \times \vec{E} = -\omega \varepsilon \vec{E} + \omega \sqrt{\varepsilon/\mu} \alpha \vec{B}. \quad (\text{S63})$$

Substituting the second equation $\vec{k} \times \vec{E} = \omega \vec{B}$ for the second term yields (1 point)

$$\vec{k} \times \vec{B} = -\omega \varepsilon \mu \vec{E}. \quad (\text{S64})$$

Altogether the equations expressed in terms of $\vec{E}$ and $\vec{B}$ read:

$$\vec{k} \cdot \vec{E} = \vec{k} \cdot \vec{B} = 0, \quad \vec{k} \times \vec{E} = \omega \vec{B}, \quad \vec{k} \times \vec{B} = -\omega \varepsilon \mu \vec{E}. \quad (\text{S65})$$

We can combine the two latter equations by taking the cross product with $\vec{k}$ and substituting the other one as well as the orthogonality. This yields the dispersion relation (1 point)

$$\omega^2 \varepsilon \mu = k^2. \quad (\text{S66})$$

We solve it for the wave velocity as (1 point; for determining v as ω/k)

$$v = \frac{\omega}{k} = \frac{1}{\sqrt{\varepsilon \mu}} = \frac{c}{\sqrt{\varepsilon_r \mu_r}}. \quad (\text{S67})$$

In the following, we consider reflection and transmission of electromagnetic waves at the flat interface ($z = 0$) of a topological insulator ($z < 0$) and free space ($z > 0$).

- b) Determine the relations between the fields $\vec{E}^\pm$ and $\vec{B}^\pm$ just above/below the interface by using the continuity equations for electromagnetic fields and the defining equations of the topological insulator. (4 points)

Solution: The continuity conditions for the electromagnetic fields require that $\vec{E}_\parallel = E_{x,y}$, $\vec{H}_\parallel = H_{x,y}$ as well as $D_\perp = D_z$ and $B_\perp = B_z$ are continuous. This yields immediately: (1 point)

$$\vec{E}_\parallel^+ = \vec{E}_\parallel^- =: \vec{E}_\parallel^\pm, \quad B_z^+ = B_z^- =: B_z^\pm. \quad (\text{S68})$$

The continuity equation $D_z^+ = D_z^-$ (1 point; for stating/using the fundamental continuity conditions in terms of D and H) together with $\vec{D}^+ = \varepsilon_0 \vec{E}^+$ and the defining relation $\vec{D}^- = \varepsilon \vec{E}^- - \sqrt{\varepsilon/\mu} \alpha \vec{B}^-$ yields (1 point; for composing with the vacuum relation and the defining relations for the topological insulator)

$$\varepsilon_0 \vec{E}_z^+ = \varepsilon E_z^- - \sqrt{\varepsilon/\mu} \alpha B_z^\pm. \quad (\text{S69})$$

The continuity equation $\vec{H}_\parallel^+ = \vec{H}_\parallel^-$ together with $\vec{H}^+ = \mu_0^{-1} \vec{B}^+$ and the defining relation $\vec{H}^- = \mu^{-1} \vec{B}^- + \sqrt{\varepsilon/\mu} \alpha \vec{E}^-$ yields (1 point; for the other quantity)

$$\mu_0^{-1} \vec{B}_\parallel^+ = \mu^{-1} \vec{B}_\parallel^- + \sqrt{\varepsilon/\mu} \alpha \vec{E}_\parallel^\pm. \quad (\text{S70})$$

An incoming plane monochromatic wave propagates in free space along the z -axis with linear polarisation along the x -axis. It hits the interface and is reflected and transmitted.

- c) Compose the fields $(\vec{E}^\pm, \vec{B}^\pm)$ at the interface in terms of the incoming, transmitted and reflected waves $(\vec{E}^{\text{I,R,T}}, \vec{B}^{\text{I,R,T}})$. Show that the electric field components at the interface satisfy the relations (5 points)

$$\begin{aligned} E_x^{\text{I}} + E_x^{\text{R}} &= E_x^{\text{T}}, & \sqrt{\mu_r/\varepsilon_r} (E_x^{\text{I}} - E_x^{\text{R}}) &= E_x^{\text{T}} - \alpha E_y^{\text{T}}, \\ E_y^{\text{R}} &= E_y^{\text{T}}, & -\sqrt{\mu_r/\varepsilon_r} E_y^{\text{R}} &= E_y^{\text{T}} + \alpha E_x^{\text{T}}. \end{aligned} \quad (9)$$

Hint: As relevant results of part a) and part b) you may assume the established relations between $\vec{E}$ and $\vec{B}$ for electromagnetic waves in matter as well as the continuity equation $\mu_r \vec{B}_\parallel^+ = \vec{B}_\parallel^- + \sqrt{\mu \varepsilon} \alpha \vec{E}_\parallel^\pm$.

Solution: The incoming wave is polarised in the x -direction, whereas he have to assume general polarisation in the x,y -plane for the reflected and transmitted waves (1 point)

$$\vec{E}^I = \vec{e}_x E_x^I, \quad \vec{E}^R = (E_x^R, E_y^R, 0), \quad \vec{E}^T = (E_x^T, E_y^T, 0). \quad (S71)$$

The waves propagate along the z -direction and we define the wave vectors as (1 point)

$$\vec{k}^I = -c^{-1}\omega\vec{e}_z, \quad \vec{k}^R = +c^{-1}\omega\vec{e}_z, \quad \vec{k}^T = -v^{-1}\omega\vec{e}_z. \quad (S72)$$

The corresponding magnetic fields are determined by $\vec{B} = \omega^{-1}\vec{k} \times \vec{E}$ which is one of the relations (S61) which also matches with the established relations for electromagnetic waves in matter. The resulting magnetic fields read (1 point)

$$\vec{B}^I = -c^{-1}E_x^I\vec{e}_y, \quad \vec{B}^R = (-c^{-1}E_y^R, c^{-1}E_x^R, 0), \quad \vec{B}^T = (v^{-1}E_y^T, -v^{-1}E_x^T, 0) \quad (S73)$$

Now the fields at the interface compose as (1 point)

$$\begin{aligned} \vec{E}^+ &= \vec{E}^I + \vec{E}^R, & \vec{B}^+ &= \vec{B}^I + \vec{B}^R, \\ \vec{E}^- &= \vec{E}^T, & \vec{B}^- &= \vec{B}^T. \end{aligned} \quad (S74)$$

For our waves, we need only the continuity relation in the parallel directions

$$\vec{E}_{\parallel}^+ = \vec{E}_{\parallel}^-, \quad v\mu_r\vec{B}_{\parallel}^+ = v\vec{B}_{\parallel}^- + \alpha\vec{E}_{\parallel}^{\pm}. \quad (S75)$$

Composing all equations, we find the desired relations (1 point)

$$\begin{aligned} E_x^I + E_x^R &= E_x^T, & \sqrt{\mu_r/\varepsilon_r}E_x^I - \sqrt{\mu_r/\varepsilon_r}E_x^R &= E_x^T - \alpha E_y^T, \\ E_y^R &= E_y^T, & -\sqrt{\mu_r/\varepsilon_r}E_y^R &= E_y^T + \alpha E_x^T, \end{aligned} \quad (S76)$$

where we note $c = 1/\sqrt{\varepsilon_0\mu_0}$ and $v = 1/\sqrt{\varepsilon\mu}$.

- d) Show that the transmitted wave is linearly polarised. Compute the angle θ_F between the incoming and transmitted electric field (Faraday rotation) and the ratio T of the intensity transmitted across the interface. (5 points)

Solution: First, we solve (9) for $E_{x,y}^R$ to obtain (1 point)

$$E_x^R = E_x^T - E_x^I, \quad E_y^R = E_y^T. \quad (S77)$$

We substitute the result back in (9) to obtain two linear equations for $E_{x,y}^T$. One of them reads

$$\alpha E_x^T + (1 + \sqrt{\mu_r/\varepsilon_r})E_y^T = 0. \quad (S78)$$

This equation shows that the resulting wave is linearly polarised because the ratio E_x^T/E_y^T is real (1 point). The equation can be solved immediately for the angle θ_F (1 point)

$$\tan \theta_F = \frac{E_y^T}{E_x^T} = -\frac{\alpha}{1 + \sqrt{\mu_r/\varepsilon_r}}. \quad (S79)$$

Finally, we solve for both components of the transmitted wave (1 point)

$$E_x^T = \frac{2\sqrt{\mu_r/\varepsilon_r}(1 + \sqrt{\mu_r/\varepsilon_r})}{(1 + \sqrt{\mu_r/\varepsilon_r})^2 + \alpha^2} E_x^I, \quad E_y^T = -\frac{2\alpha\sqrt{\mu_r/\varepsilon_r}}{(1 + \sqrt{\mu_r/\varepsilon_r})^2 + \alpha^2} E_x^I. \quad (S80)$$

The average intensities of the initial and transmitted wave take the form (1 point)

$$\begin{aligned} \langle \vec{S}^I \rangle &= \mu_0^{-1} \langle \vec{E}^I \times \vec{B}^I \rangle = -\frac{1}{2}\sqrt{\varepsilon_0/\mu_0} (E_x^I)^2 \vec{e}_z, \\ \langle \vec{S}^T \rangle &= \mu^{-1} \langle \vec{E}^T \times \vec{B}^T \rangle = -\frac{1}{2}\sqrt{\varepsilon/\mu} ((E_x^T)^2 + (E_y^T)^2) \vec{e}_z. \end{aligned} \quad (S81)$$

For the ratio we obtain (1 point*; if missing out on one of the above points)

$$T = \frac{\langle \vec{S}_z^T \rangle}{\langle \vec{S}_z^I \rangle} = \sqrt{\varepsilon_r/\mu_r} \frac{(E_x^T)^2 + (E_y^T)^2}{(E_x^I)^2} = \frac{4\sqrt{\mu_r/\varepsilon_r}}{(1 + \sqrt{\mu_r/\varepsilon_r})^2 + \alpha^2} \quad (S82)$$